

Answers

1. What do you get by subtracting 1 point from another?
Answer: *You get a direction vector that represents the change or difference between the 2 points. In essence, it determines the change needed to go from the first point to the 2nd.*
2. What do you get by adding 2 vectors?
Answer: *When you add 2 vectors, the result is another vector.*
3. What do you get by adding 2 points?
Answer: *Adding points doesn't really make much sense in Linear Algebra. Although, in some specific cases, such as finding the midpoint between 2 points, you may still do it.*
4. What do you get by adding a point and a vector?
Answer: *When you add a point and a vector, you get another point.*
5. What do you get by subtracting a vector from a point?
Answer: *When you subtract a vector from a point, you get another point.*
6. In a 1D space, if P1 is located at 4, and there is a direction vector of +7, where will the new point be?
Answer: $4 + 7 = 11$
7. The temperature today at 9 am is 0° . By noon, it rises by 8° . What is the temperature at noon?
Answer: $0^{\circ} + 8^{\circ} = 8^{\circ}$
8. Given two vectors: +6 and -3, what is their sum?
Answer: $6 + -3 = +3$
9. Consider a point P at position 12. Add a direction vector of +5 to it. Where is the new point?
Answer: $12 + 5 = 17$
10. If the temperature rises by 5° in the afternoon and then drops by 2° in the evening, what is the total change in temperature?
Answer: $5^{\circ} + -2^{\circ} = 3^{\circ}$
11. Starting at point 0, apply the vectors +4, -7, and +3 sequentially. Where will you end up?
Answer: $0 + 4 + -7 + 3 = 0$
12. If the temperature at 8 am is 7° and at 5 pm is 15° , what was the change in temperature?
Answer: $15^{\circ} - 7^{\circ} = +8^{\circ}$

13. Point A is at 20 and Point B is at 35. What is the direction vector from A to B?

Answer: $B - A = 35 - 20 = +15$

14. Given two points: 25 and 18, find the direction vector from the first point to the second.

Answer: $18 - 25 = -7$

15. Point A is at 6. It moves according to the direction vector +9. After that, it follows another direction vector of -4. Where is Point A now?

Answer: $A + 9 + -4 = 6 + 9 - 4 = 11$

16. Point X is at 10. Point Y is at 30. Find the direction vector from X to Y and then subtract a vector of -5. What's the resultant vector?

Answer: $(Y - X) - 5 = (30 - 10) - 5 = 15$

17. If Point P1 is at 7 and Point P2 is at 19, calculate the halfway point between them using the method we covered.

Answer: $7 + 0.5 * (19 - 7) = 13$

18. Calculate the halfway point between points 15 and 35.

Answer: 25

19. If the temperature changed from -5° to 10° between Monday to Tuesday and then from 10° to -3° from Tuesday to Wednesday, how much did the temperature change in total over the two days?

Answer:

Monday to Tuesday: $10^{\circ} - -5^{\circ} = +15^{\circ}$

Tuesday to Wednesday: $-3^{\circ} - 10^{\circ} = -13^{\circ}$

Over the two days: $15^{\circ} + -13^{\circ} = 2^{\circ}$

20. Given that Point A is at 12, after applying vectors +7, +3, and -8 in sequence:

a. Where is Point A?

b. What is the combined vector of these changes?

Answers:

a. Vector: $7 + 3 - 8 = 2$

b. Position: $12 + 2 = 14$